

CONSTRUCT1

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Problem

With an unmarked, infinite two-sided ruler, construct the midpoint of a given segment.

Solution. Name those points to be A and B . Construct a line passing through A and B . Consider any point C on the plane.

Since the ruler is two-sided, place one side along AB and use the other side to draw a line parallel to AB . Say that intersects CA and CB at E, F respectively. Let $X = BE \cap AF$.

Due to Thales' Theorem and Ceva's Theorem, it is well known that CX intersects AB at its midpoint and as a result we are done.